Methodology comparison: independent models of XUV-driven atmospheric escape, and what EXHALE took from each

1. Purpose, and the verdict first

This memo compares the methodology of four independent modelling efforts for XUV-driven atmospheric escape from close-in exoplanets: the general-purpose 1-D radiation-hydrodynamics code AIOLOS (Schulik & Booth 2023, MNRAS 523, 286); the dedicated thermosphere–ionosphere escape model of Taylor et al. (2025, 2026), built on the Koskinen et al. code; the multi-fluid PLUTO-based model of Xing et al. (2023); and the photochemical–hydrodynamic model of Frelikh & Murray-Clay (2026), and then records, item by item, what EXHALE has taken from each and what it has not. The memo was first written as a forward-looking recommendation list. It is now a record of outcomes, so every candidate carries one of three verdicts: **adopted** (and in which module), **adopted then replaced** (and why the first choice did not survive), or **not adopted** (and whether the reason still holds). Of the fifteen items assessed in the adoption assessment below, six are adopted, two were adopted and then replaced, and seven are not adopted. The two largest gaps the original assessment identified (no radius-dependent He/H separation, and an ad hoc lower boundary) are both closed, and closed further than the recommendation asked for.

Frelikh & Murray-Clay is the newest of the four and the closest peer to EXHALE's molecular layer, so it is the source of four of the seven not-adopted items and of one finding that belongs to neither code alone: the thermal-dissociation rate pair that both use is not thermodynamically consistent below about 3000 K (Section 7.). Three further points where the two codes diverge, and where the judgement went against EXHALE, are the photoelectron partition in a molecular gas (Section 6.6), the absence of a composition boundary condition at the base (Section 6.9), and the sharpness of the modeled $H_2 \rightarrow H$ transition (Section 6.10).

Every statement about EXHALE below was checked against `src/` rather than against EXHALE's own documentation; the file and routine names given are the ones in the tree. The AIOLOS, Taylor, Xing and Frelikh entries are what those papers say. Numbers measured here from EXHALE output, rather than quoted from a paper, are marked as such at the point of use.

2. The codes considered

- **AIOLOS**: the general-purpose 1-D multi-species radiation-hydrodynamics code of Schulik & Booth (2023, MNRAS 523, 286; "SB23"), distributed at `github.com/Schulik/aiolos`.
- **Taylor et al. (2025)**, *ApJ* 989, 68: "A Multispecies Atmospheric Escape Model with Excited Hydrogen and Helium: Application to HD209458b," built on the Koskinen et al. (2013a,b; 2022) thermosphere–ionosphere code.
- **Taylor et al. (2026)**, *ApJ* 999, 214: "Helium Escape in Context: Comparative Signatures of Four Close-in Exoplanets," a direct application and extension of Taylor (2025) to HD 209458 b, HD 189733 b, HD 149026 b and

GJ 1214 b.

- **Xing et al. (2023)**, *ApJ* 953, 166: "The Mass Fractionation of Helium in the Escaping Atmosphere of HD 209458b," a *multi-fluid* model built inside PLUTO.
- **Frelikh & Murray-Clay (2026)**, *ApJ* 996, 96: "Efficiency of Hydrodynamic Atmospheric Escape in Hot Jupiters and Super-Earths," a 1-D photochemical–hydrodynamic model of H–He escape with H_3^+ and Lyα cooling, heat conduction, tidal gravity and photoelectron secondary ionization.

"Taylor" denotes the shared Koskinen-based framework used in both Taylor papers unless a specific paper is named. "Frelikh" denotes Frelikh & Murray-Clay (2026).

3. One-paragraph characterization of each code

AIOLOS. A *general-purpose* radiation-hydrodynamics code, not a dedicated escape code. It solves the time-dependent Euler equations (mass, momentum, energy) for an arbitrary number of species (each a **separate fluid** with its own density, momentum and internal energy) coupled by an inter-species **friction/drag** solver. It uses finite-volume **HLLC** fluxes with **PLM** reconstruction (monotonized-central limiter) and second-order **strong-stability-preserving** Runge–Kutta time stepping, in Cartesian, cylindrical or **spherical** geometry. Radiation is treated with **flux-limited diffusion (FLD)** over arbitrary in/out spectral bands, solved implicitly through a block-tridiagonal system. Escape is one of many problems it can run.

Two things about AIOLOS have to be attributed to the distributed code rather than to SB23, because the paper does not carry them. The paper implements a C2Ray-style two-species photoionization scheme and calls a fuller chemical solver future work; the code has since acquired a reaction-network class (`c_reaction`) with selectable photochemistry levels, the Black (1981) heating and cooling rates with free-free and H_3^+ cooling (`photochem.cpp`, `chemistry.cpp`), grid spacing that can be linear, logarithmic or bi-logarithmic (`PARI_GRID_TYPE`), and a tidal-field option (`USE_TIDES`). Statements below about AIOLOS's chemistry, cooling, grid or tides refer to the code.

Taylor (2025/2026). A *dedicated* 1-D spherical thermosphere–ionosphere and hydrodynamic-escape model (Koskinen et al. 2013/2022), integrated to steady state, with a **single bulk momentum equation** but **multi-species molecular and eddy diffusion** that redistributes species, so He and H can separate diffusively. Its distinguishing strengths are in the completeness of the *microphysics*: a full He(2^3S) and H(n=2) non-LTE network solved by the KPP kinetic pre-processor, a **Lyα Monte Carlo** radiative-transfer model (Huang et al. 2017) iterated with the hydro solution for Hα, a high-resolution He(2^3S) photoionization cross section (B-spline K-matrix), temperature-dependent Penning ionization, and self-consistent coupling to a photochemical lower and middle atmosphere at the μbar level.

Xing (2023). A *dedicated* 1-D spherical escape model whose distinguishing feature is that it is genuinely **multi-fluid**: H, H^+, He, He^+ and e^- each carry their **own velocity**, coupled by resonant and non-resonant collisional drag. It is the only one of the three that captures **He/H mass fractionation** (heavy He under-dragged by escaping H) in the dynamics itself rather than through a diffusion coefficient. It is built on PLUTO (HLL Riemann solver, third-order TVD RK) with a Riemann split that carries the electron/electric-field pressure term. The He(2^3S) 10830 Å line is post-processed,

not solved in the hydro loop.

Frelikh (2026). A *dedicated* 1-D spherical escape model built around a molecular base, and of the four the closest peer to EXHALE's lower atmosphere. It is **single-fluid** in velocity (all species travel with the bulk u, and the diffusion the code contains is turned off for the published H–He runs), but it carries a mass-conservation equation for each of the eight H and He species with a chemical source term. The hydrodynamics are unusual for this field: no Riemann problem at all, but the **constrained interpolation profile** (CIP) advection scheme of Yabe & Aoki (1991) on a **staggered mesh**, operator-split against explicitly differenced source terms, with a Landshoff-plus-von Neumann artificial viscosity for stability. The 22-reaction H–He network (their Table 1, compiled from Yelle 2004 and García-Muñoz 2007) is **integrated in time** as a stiff ODE system by the semi-implicit Euler stepper `StepperSie` of Press et al. (2007) with an **analytic Jacobian**. Radiative cooling is Lyα (Black 1981) and H_3^+ (Miller et al. 2013); heat conduction follows García-Muñoz (2007). Photoelectron secondary ionization uses Dalgarno et al. (1999), including the H_2 vibrational, triplet-dissociation and Lyman–Werner channels. The published result is a three-layer atmosphere: an H_3^+-cooled H_2 layer at the base, a Lyα-cooled neutral-H layer above it, and a PdV-cooled ionized outflow. There is no He metastable, no He↔H charge exchange and no metal in the network.

EXHALE. A 1-D spherical **single-fluid** escape code of ATES heritage. The hydrodynamics are finite-volume, second-order, with PLM or WENO3 reconstruction (usable as a two-stage PLM→WENO3 sequence) and a selectable LLF/**HLLC**/Roe numerical flux (`flux/Num_Fluxes.f90`); reconstructed face states are dropped to the piecewise-constant limit wherever they lose positivity (`positivity_limited_faces`, `states/Reconstruction.f90`), and a gated fourth-difference dissipation restores the damping of the contact mode that HLLC loses as the flow stalls (`flux/low_mach_dissipation.f90`, key `Low-Mach damping`, off by default). The steady wind is reached either by explicit marching on the mass-flux ($\rho v r^2$) spread or by a direct **Jacobian-free Newton–Krylov** solve: right-preconditioned GMRES(m) with pseudo-transient continuation (`time_step/steady_newton.f90`, key `Solver: Newton`). Ionization of H, He, ten metal elements in 27 ion stages, the He(2^3S) metastable and, when molecular chemistry is on, $H_2/H_2^+/H_3^+/HeH^+$ is solved as *one* coupled nonlinear algebraic system (analytic-Jacobian Newton, MINPACK `hybrd1` fallback; the `System_HeH_*` family). The photoelectron heating fraction is not a free parameter: it follows the Shull & van Steenberg (1985) branching evaluated on each cell's own ionized fraction and neutral composition. Helium is transported as the second component of a binary mixture with stage-resolved friction (`functions/binary_element_diffusion.f90`), the Navier–Stokes viscous force and heat conduction are available (`time_step/viscous_conduction.f90`), and the base can be handed over from a photochemical lower atmosphere as scalars or as a full profile. Diagnostics are a non-LTE He(2^3S) network, a non-LTE H(n=2) population (Christie et al. 2013), Lyα transfer through a Neufeld escape-probability closure or an imported field, and the transmission post-processor `EXHALE_transit.py`. Size of the production build: 95 Fortran source files, 50 762 SLOC excluding blank and comment lines (`src/utils/codesize.py`, measured 2026-09-11), of which 30 003 are in modules that do not exist in upstream ATES; the appendix of `Update_EXHALE.pdf` carries the full table.

4. Side-by-side methodology table

A sixth column does not fit across one A4 text block, so the comparison is split by axis: Table 1 carries the numerical method and Table 2 the physics and the diagnostics.

Table 1: Numerical method. The four external columns are as described in their papers; the EXHALE entries were verified in `EXHALE_v1.00/src/`.

Axis	AIOLOS (SB23)	Taylor 2025/2026	Xing 2023	Frelikh 2026	EXHALE
Purpose	General multi-species RHD	Dedicated escape + line diagnostics	Dedicated escape, He/H fractionation	Dedicated escape; radiative-cooling efficiency	Dedicated escape + line diagnostics
Dimensions	1-D (sph./cyl./cart.)	1-D spherical	1-D spherical	1-D spherical, substellar ray	1-D spherical
Route to steady state	Time-dependent → steady	Time-dependent → steady	Time-dependent → steady	Time-dependent → steady	RK marching on the $\rho v r^2$ spread, optionally finished by a direct JFNK solve
Fluid model	**Multi-fluid** + friction	**Single bulk** + multi-species diffusion	**Multi-fluid**	**Single fluid**; one mass equation per species, all at the bulk u	**Single fluid** + binary element diffusion
Advection scheme	PLM + MC limiter	Flux-conservative van Leer advection (Koskinen et al. 2013a); not restated in the Taylor papers	PLM (PLUTO)	**CIP** cubic interpolation carrying X and $\partial_r X$, on a staggered mesh	PLM, WENO3, or two-stage PLM→WENO3, with a face positivity guard
Riemann solver	HLLC	-	HLL (+ e^--pressure split)	**None**; Landshoff + von Neumann artificial viscosity	LLF, HLLC or Roe (`Numerical flux`)
Time integrator	Second-order SSP RK	Operator split, semi-implicit Crank–Nicholson, forward to steady	Third-order TVD RK	Operator split: CIP advection, then explicitly differenced source terms	RK marching; GMRES(m) JFNK with PTC for the steady solve
Grid	Linear, log or bi-log (code)	Radial	1024 points, stretched, 1–10 R_p	Nonuniform, staggered, 1–8.8 R_p, ghost cells	Uniform, stretched or mixed; cell count set at runtime

Table 2: Physics and diagnostics. Conventions as in Table 1. EXHALE options marked *opt-in* default to off.

Axis	AIOLOS (SB23)	Taylor 2025/2026	Xing 2023	Frelikh 2026	EXHALE
Radiation	**FLD**, multi-band, implicit	XUV attenuation + Lyα Monte Carlo	Radial XUV attenuation, 53 bins	Radial XUV attenuation, 37 bins (Richards et al. 1994), 12–248 eV	Radial XUV attenuation on a multi-bin energy grid (power law or tabulated SED) + Lyα transfer
Photoelectron heating	From the C2Ray energy split	Tuned efficiency, 20–40 %	Tuned frequency-averaged η, 0.1–0.5	**Dalgarno et al. (1999)** $W_0(1+Cx^\alpha)$, with the H_2 vibrational, triplet and Lyman–Werner returns added back	Shull & van Steenberg (1985) branching renormalized on the cell's own neutral composition; **no efficiency parameter**

Table 2, continued

Axis	AIOLOS	Taylor	Xing	Frelikh	EXHALE
Chemistry solve	C2Ray two-species (paper); reaction network (code)	KPP preprocessor over a stiff ODE network	Semi-implicit source terms	**Stiff ODE integrated in time**: semi-implicit Euler (`StepperSie`), **analytic** Jacobian, LU	**One coupled nonlinear algebraic system** per cell; analytic-Jacobian Newton, or MINPACK `hybrd` under a continuation
He/H separation	Via friction/drag	Molecular + eddy diffusion of the trace ratio $n_{\rm He}/n_{\rm H}$	Via multi-fluid dynamics	Implemented but turned off for the published H–He runs	Binary diffusion of the mass fraction $X{=}\rho_{\rm He}/\rho$, friction resolved by ionization stage; *opt-in*
Metals	Through the code's chemistry	Optional (solar)	None	**None**	**10 elements, 27 ion stages** in the coupled system, with CHIANTI closed-form line cooling
Molecules	Through the code's chemistry	H_2, H_2^+, H_3^+, HeH^+ (2026, for HD 189733 b and GJ 1214 b only)	None	Same four species; 22 reactions (their Table 1)	Same four species inside the coupled system, with Lyman–Werner photodissociation; *opt-in*
He↔H charge exchange	Through the code's chemistry	Yes	Yes	**Absent from the network**	Huang et al. (2023) Table 4, rows B1/B2: a non-radiative and a radiative channel, separately
Cooling	Recombination + Lyα (paper); Black (1981) line/free-free + H_3^+ (code)	Recombination + H I lines + H_3^+	**Lyα only**	Lyα + H_3^+ only; recombination cooling checked and left off	Lyα, recombination, free-free, CHIANTI metal lines with fine-structure line trapping, H_3^+ infrared
H_3^+ cooling form	Black (1981) (code)	Miller et al. (2013)	-	Miller et al. (2013), LTE, emission into vacuum; zero above 5000 K	MiZATeP cooling function (Mizus et al. 2017; the Miller et al. 2013 LTE fit until 2026-10-10) with the Miller et al. (2013) non-LTE departure factor; net exchange with a diluted base field, *opt-in*
He(2^3S) 10830	Not a focus	In-loop non-LTE	Post-processed (Yan et al. 2022)	**Not in the network**	In-loop non-LTE (default on) + `EXHALE_transit.py`
H(n=2)/Hα	Not a focus	Non-LTE + Lyα Monte Carlo	Post-processed	Not computed	Non-LTE (Christie et al. 2013) + Lyα escape probability or an imported field
Viscosity / conduction	-	Molecular + eddy diffusion, conduction	Collisional drag	Heat conduction (García-Muñoz 2007); no viscosity	Navier–Stokes viscosity and heat conduction (Crank–Nicolson); *opt-in*
Tidal/Roche	`USE_TIDES` (code)	Tested (2025); off (2026)	Stellar tidal term	Tidal term (their Appendix C)	Roche potential, spherical or Roche-lobe domain

Table 2, continued

Axis	AIOLOS	Taylor	Xing	Frelikh	EXHALE
Lower boundary	Configurable	**μbar, coupled to photochemistry**	1 R_p, fixed n, T = 1500 K	$\approx$1 μbar: **species densities and p imposed**, u extrapolated; Newtonian relaxation to T_0 below τ=3	μbar base: fixed T,n; or an analytic column; or a scalar `base.inp`; or a full photochemical profile with $K_{zz}(p)$
Transmission	-	Ray-traced Voigt	Post-processed 10830	H_3^+ emission spectrum (ExoCross); no transit spectrum	`EXHALE_transit.py`: 10830, Lyα, Hα, Hβ, metal doublets, optional triaxial Roche geometry

5. What they have in common

1. **1-D spherical XUV-driven escape**, with photoionization heating balanced by Lyα/recombination cooling and adiabatic expansion, is the shared physical backbone.
2. **A steady transonic wind is the target.** The four external codes evolve the time-dependent equations forward; EXHALE marches in the same way but can also solve the steady equations directly. The physical end state is the same.
3. **Much of the same core H/He atomic physics.** Voronov collisional ionization, radiative and dielectronic recombination, and H$\leftrightarrow$He charge exchange appear in Xing, Taylor and EXHALE with near-identical rate coefficients, differing mainly in the source table (Storey & Hummer, Badnell, Verner). EXHALE's charge-exchange network is now Huang et al. (2023) Table 4 (`radiation/charge_exchange.f90`). Frelikh is the exception on the last of these: its Table 1 has no He^++H and no He+H^+ reaction of any kind, and neither does its text (Section 6.4).
4. **He(2^3S) as the key observable.** Taylor, Xing and EXHALE all target the 10830 Å line through a non-LTE 2^3S population (recombination, collisions, radiative decay, Penning ionization, photoionization). AIOLOS and Frelikh do not; the He metastable is not a species in either.
5. **Finite-volume, slope-limited, shock-capturing hydrodynamics** underlies AIOLOS, Xing (PLUTO) and EXHALE. Two of the five are outliers, and in the same direction: the Koskinen solver beneath Taylor is flux-conservative van Leer advection operator-split against a semi-implicit Crank–Nicholson step (Koskinen et al. 2013a), and Frelikh is CIP advection operator-split against explicitly differenced source terms. Both are staggered, both use an added dissipation in place of an upwind flux, and **neither poses a Riemann problem**. Neither Taylor paper restates its numerical scheme; Frelikh gives its scheme in full in its Appendix A.

One item that used to be on this list no longer belongs. It read: none of the four derives the photoelectron heating efficiency from first principles, and all treat it as the primary free knob. That is no longer true of EXHALE. Its heating rate is the integral of the excess photon energy over the energy grid weighted by the Shull & van Steenberg (1985) heat fraction, and that fraction is a function of the local ionized fraction with the ionization branching renormalized onto the cell's own neutral H and He (`svs85_secondary_branching`, `radiation/util_ion_eq.f90`); there is no heating-efficiency key in `input.inp` at all. Taylor and Xing still tune theirs, and AIOLOS takes its split from C2Ray.

6. Where they genuinely differ

6.1 Fluid model

- **Xing is multi-fluid**, each species carrying its own velocity, which is what reproduces **He/H mass fractionation**: because He is four times heavier it is under-dragged by the H wind, so the *escaping* He/H is about half the bulk abundance, a weak 10830 line without a subsolar He abundance.
- **Taylor keeps a single bulk momentum equation and adds molecular and eddy diffusion**, which produces the same qualitative diffusive separation at far lower cost. Two numbers are quoted in Taylor (2025) and they belong to different models: the best-fit model's elemental He/H falls from 8 % at the base of the thermosphere to about 2.5 % at high altitude, while the model Taylor sets against Xing's and Schulik & Owen's (2025) multi-fluid results is Model B, whose H/He goes from 92/8 at the base to about 96/4.
- **AIOLOS is multi-fluid but escape-agnostic**: its friction solver would in principle give fractionation, but the code is not tuned for the transonic-wind problem.
- **EXHALE keeps one bulk velocity and diffuses the composition on top of it**, which is Taylor's choice of architecture but not Taylor's variable. Where Taylor transports the trace ratio $n_{\mathrm{He}}/n_{\mathrm{H}}$, the binary operator (`binary_element_diffusion.f90`) transports the helium *mass fraction* $X = \rho_{\mathrm{He}}/\rho$ of a two-component mixture, which is bounded at both ends of the composition axis and therefore does not break down as helium approaches unity. Hydrogen's carriers are H I, H II, H_2, H_2^+ and H_3^+, helium's are He I, He II and He III, and the binary coefficient is a Blanc's-law mixture over their stage fractions, with hard-sphere (Banks & Kockarts 1973), polarization ion–neutral (Schunk & Nagy) and Coulomb ion–ion (Paquette et al. 1986) pair coefficients chosen by charge. That last point is the physical content: an ion is held to the protons by Coulomb friction instead of settling at a neutral rate.

6.2 Radiation transport

AIOLOS is the only one with a full **FLD** thermal-radiation solver, which is necessary for optically-thick interiors and dust and overkill for the escaping thermosphere. Taylor is the only one doing **Lyα Monte Carlo** transfer iterated with the hydro solution. Xing and EXHALE both use radial XUV attenuation: EXHALE on an energy grid built per run from the ionization thresholds, driven by either a power-law XUV spectrum or a tabulated SED. On top of that EXHALE closes Lyα trapping either with a the escape probability of the static plane-parallel damping-wing slab (Neufeld 1990 eq. 3.27, Harrington 1973 eq. 40) evaluated in line every step, or with an externally computed $J_{\mathrm{Ly}\alpha}(r)$ profile read from file (`radiation/lya_rt.f90`; keys `Jlya escape-prob` and `Jlya RT file`). The second of those is how a real Monte Carlo field enters: the LaRT scattering-rate field is produced offline (`examples/exhale_to_lart.py`) and read back in.

6.3 Ionization and chemistry solver

Taylor uses the KPP kinetic preprocessor over a large stiff ODE network; AIOLOS uses the C2Ray two-species scheme of SB23, or the reaction network its code has since grown; Xing folds semi-implicit source terms into the PLUTO

RK stages. EXHALE solves a single coupled *nonlinear algebraic* system per cell, which is what a steady-state code needs, and everything lives in it together: H, He, the He(2^3S) metastable, the ten metal elements, and the four molecular species. Adding physics means adding rows to that system rather than a parallel decoupled solver. The molecular rows are normalized by a turnover-rate scale before they reach the solver, because the He and H blocks otherwise differ in magnitude by six orders at He/H = 10^3 and `hybrd1` does not scale rows itself.

Frelikh integrates the network in time; EXHALE solves it for equilibrium. This is the sharpest methodological split in this memo, and it is worth stating precisely because the two codes model the same layer. Frelikh splits the fluid equations into an advective part and source terms, and solves the chemistry in a substep of its own: of Equation (6) they write that the chemical source term is one "which we solve for in a separate substep of our routine." The network is then a stiff initial-value problem, and their Section 2.9 says so:

> "As the rates of the reactions vary by several orders of magnitude, the reaction network cannot be solved via a straightforward explicit method—the set of equations is said to be 'stiff.' We implement the semi-implicit method of solving stiff systems of equations from W. H. Press et al. (2007), the details of which are given in Appendix B."

Their Appendix B gives the whole of it: semi-implicit Euler as in Numerical Recipes Section 17.5.2, the specific stepper named (`StepperSie`), the linear system solved by LU decomposition with forward and back substitution, Δt crossed by a sequence of substeps dt_i with Bulirsch–Stoer polynomial extrapolation to test each one, and the tolerances quoted (`rtol` = 10^{-7}, `atol` = 10^{-4} `rtol`, $h_1 = 10^{-6}$, `hmin` = 0). The Jacobian is analytic: "The Jacobian $\partial f/\partial y$ is calculated analytically for our problem, as are the time derivatives of the concentrations on the left-hand side of Equation (B6)." Cells are independent, so the network runs in parallel across the grid.

Set against that, EXHALE has the more careful *formulation* and the weaker *linear algebra*. Its rows are steady-state production–loss balances (`System_HeH_mol.f90`); there is no path in the tree that integrates the network in time. `nonlinear_system_solver/constrained_chemical_equilibrium.f90` takes the unknowns to be $u_k = \ln n_k$, so no iterate can go negative, following the CEA method of Gordon & McBride (1994); it writes element conservation as an explicit residual row per element rather than as a closure, so that HeH^+ satisfies the H and the He budget simultaneously; and it takes n_e from charge neutrality rather than carrying it as an unknown, so neutrality holds identically at every iterate. Each rung is then handed to MINPACK `hybrd` with a *forward-difference* Jacobian: the opposite of Frelikh's choice.

The reason EXHALE needs the extra apparatus is stated in that file's own header: the molecular cell is bistable, with "a molecular basin in the dense, shielded base and an atomic basin above the $H_2 \rightarrow H$ front," so a single starting guess either lands in the right basin or does not. The fix is a natural-parameter continuation in the radiation field, tracked from six decades below the cell's own field (where the composition is known in closed form) up to the true field, bisecting any rung that fails.

The judgement worth recording is that Frelikh does not meet this problem at all, and not because its solver is better. A time-integrated network has no uniqueness question in it: the basin is selected by the previous step's state, not by a guess. Their only defense against a composition leaving the physical domain is a hard floor, stated in their Section 2.5 as "We implement a number density floor of zero for all species," and they record that the floor leaves a visible mark: their Figure 4 caption attributes a kink in the H_2 density profile to "applying the density floor in a

region where the profile drops off steeply." EXHALE's continuation is, in effect, a reconstruction inside an equilibrium formulation of the path that time integration supplies for free. Neither approach is wrong; the point is that the bistability is a cost of choosing equilibrium, not a fact about the physics.

6.4 The molecular reaction network

Frelikh's Table 1 and EXHALE's `lower_atmosphere/mol_rates.f90` are the only two molecular H–He networks among the five that can be compared reaction by reaction, and they largely agree. Frelikh lists 22 reactions compiled from Yelle (2004) and García-Muñoz (2007); EXHALE carries 23 transcribed from Koskinen et al. (2022) Table 1, which was checked here against the published table rather than against the transcription. All 22 of Frelikh's arrows are one-directional; there is no reversible pair notation in the table.

Twelve are identical in value and in source: H^+ and He^+ radiative recombination (Storey & Hummer 1995), H_2^+ dissociative recombination (Auerbach et al. 1977), H_2^++H_2 (Theard & Huntress 1974), H_2^++H (Karpas et al. 1979), H^++H_2 (Yelle 2004), three-body H_2 formation $8.0\times10^{-33}(300/T)^{0.6}$ (Ham et al. 1970), He^++$H_2 \rightarrow HeH^+$ (Schauer et al. 1989), HeH^++H_2 (Bohme et al. 1980), HeH^++H (Karpas et al. 1979), HeH^+ dissociative recombination (Yousif & Mitchell 1989), and the three-body H_3^+ association. The last of these is worth naming because it is easy to assume it is missing: Frelikh's **k21**, H^++H_2+M $\rightarrow$ H_3^++M at 3.2×10^{-29} cm^6 s^{-1} (Miller et al. 1968), **is present in EXHALE** as R13, `rk_R13_Hp_H2_M`, at the same value and from the same source.

Five differ, and Table 3 gives the sizes. Four are choices of source table. The fifth is not.

Table 3: Rate coefficients that differ between Frelikh's Table 1 and EXHALE's transcription of Koskinen et al. (2022) Table 1. Ratios evaluated here.

Reaction	Frelikh	EXHALE	Ratio
H_3^++e$\rightarrow$$H_2$+H	$2.9\times10^{-8}(300/T)^{0.65}$ (Sundström 1994)	$2.16\times10^{-8}(300/T)^{0.65}$ (Larsson 2008)	1.34
H_3^++e$\rightarrow$3H	$8.6\times10^{-8}(300/T)^{0.65}$ (Datz 1995)	$5.04\times10^{-8}(300/T)^{0.65}$ (Larsson 2008)	1.71
H_3^+ *recombination, total*	$1.15\times10^{-7}(300/T)^{0.65}$	$7.20\times10^{-8}(300/T)^{0.65}$	**1.60 at every** T
H_3^++H$\rightarrow$$H_2^+$+$H_2$	2.0×10^{-9}, **no barrier** (Yelle 2004)	$2.1\times10^{-9}e^{-20000/T}$ (Harada 2010)	4.6×10^8 (1000 K); 4.6×10^6 (1300 K); 2.1×10^4 (2000 K); 748 (3000 K)
H_2+M$\rightarrow$H+H+M	$1.5\times10^{-9}e^{-4.8\times10^4/T}$	$k_{\rm rec}/K_{\rm eq}$, detailed balance of R15 (*since 2026-09-01; was* $1.5\times10^{-9}e^{-48350/T}$*, which both codes cite from Baulch et al. 1992*)	EXHALE/Frelikh = 0.079 (1000 K); 0.19 (1300 K); 0.51 (2000 K); 0.87 (3000 K). Was 0.705 (1000 K); 0.839 (2000 K)
He^++$H_2$$\rightarrow$$H^+$+H+He	8.8×10^{-14} (Schauer 1989)	$1.0\times10^{-9}e^{-5700/T}$ (Moses & Bass 2000)	38 (1000 K); 142 (1300 K); 657 (2000 K)

H_3^++H is a physics error in Frelikh, not a source choice. The reaction $H_3^+ + H \rightarrow H_2^+ + H_2$ is endothermic. From the standard enthalpies of formation (1488.3 kJ mol^{-1} for H_2^+ (which is IE(H_2) = 15.426 eV), 1107 for H_3^+ and 218.0 for H) the reaction enthalpy is $\Delta H = 1488.3 - 1107 - 218.0 = 163.3$ kJ mol^{-1} = 1.69 eV, i.e. a barrier of 19 600 K. The Harada et al. (2010) form that EXHALE carries, $e^{-20000/T}$, matches that endothermicity; a temperature-independent 2.0×10^{-9} does not, and there is no v-excited label on the entry as there is on the neighbouring $H^+ + H_2(v \geq 4)$ row. **The judgement is that EXHALE is right here and Frelikh is not**, at the four to eight orders of magnitude the table gives for the molecular layer. The caveat is that Yelle (2004) is not in `references/`, so whether the printed constant faithfully transcribes its source could not be checked.

Nothing in Frelikh's network is absent from EXHALE. Its k1, k3, k4, k14 and k22 are photoreactions, and EXHALE has H, He and H_2 photoionization (`util_ion_eq.f90`) and Lyman–Werner photodissociation (`lyman_werner.f90`, opt-in) of its own. The one unchecked item is whether EXHALE implements the dissociative photoionization channel $H_2 + h\nu \rightarrow H + H^+ + e^-$ separately.

Six EXHALE reactions have no counterpart in Frelikh: collisional ionization of H and of He (Voronov 1997), electron-impact dissociation of H_2 (Stibbe & Tennyson 1999), H_2+He^+ (Barlow 1984), and the two He↔H charge-exchange channels. The last of these is the one that matters. Frelikh's seven helium reactions (k14–k20) cover photoionization, two He^++H_2 channels, two HeH^+ channels, He^+ recombination and HeH^+ dissociative recombination, and **contain no charge exchange with hydrogen in either direction**. In EXHALE the live definitions are in `radiation/charge_exchange.f90`, Group B, and the two are not each other's reverse: B1, $He + H^+ \rightarrow He^+ + H$, is the non-radiative collisional channel (Kimura et al. 1993), while B2, $He^+ + H \rightarrow He + H^+ + \text{photon}$, is radiative charge transfer (Stancil et al. 1998). A photon-emitting process has no collisional reverse, which is why the module forbids testing the pair against detailed balance.

6.5 H_3^+ infrared cooling

Frelikh takes the cooling function from Miller, Stallard, Tennyson & Melin (2013), and so did EXHALE until 2026-10-10 (its 800–1800 K coefficients were the same seven numbers as Frelikh's Equation (13)). EXHALE now takes the LTE emission from the MiZATeP cooling function of Mizus et al. (2017, MNRAS 468, 1717), tabulated at 1 K from 20 to 5000 K: the Miller fit lies above it by 1.10, 1.48, 1.89 and 2.1 at 900, 1500, 2000 and 3000 K, and the NMT line list Miller et al. state they summed agrees with MiZATeP to 1 % over 300–2000 K (update log stage 3 section 132; `md/h3p_cooling_cause_20261010.md`). So at the temperatures of Frelikh's wind the two codes now differ in the cooling function itself. Three further things differ.

First, **Frelikh radiates into vacuum and EXHALE can exchange with a field**. Frelikh's volumetric rate is $\Lambda_{H_3^+} = 4\pi E(T) n_{H_3^+} \times 10^7$ erg cm^{-3} s^{-1}, with no incident term, no stimulated emission and no optical-depth factor; downward emission is treated as lost: "We treat the radiation that is emitted downward (i.e., toward the lower simulation boundary) as having left the domain." Optical thinness is verified after the fact in their Section 3.4 against ExoCross cross sections. EXHALE's `lower_atmosphere/h3p_cooling.f90` carries the same emission term but its `h3p_net_cooling_rate` returns $\Lambda_{net} = n_{H_3^+} 4\pi\, s(T, n_{H_2}) [E(T) - W_{dil} E(T_{rad})] \times 10^7$, which is exactly zero when $T = T_{rad}$ and $W_{dil} = 1$. **But W_{dil} is reached only through the key `Base IR field`, which defaults to off**, so EXHALE's default behavior is Frelikh's.

Second, EXHALE carries the **non-LTE departure factor** $s(T, n_{H_2})$ of Miller et al.'s Table 6 and an LTE emission

defined from 20 to 5000 K (the MiZATeP table; before 2026-10-10 all four published Miller segments, 30–300, 300–800, 800–1800, 1800–5000 K); Frelikh is pure LTE and implements only the upper two Miller segments, so its H_3^+ cooling is undefined below 800 K and is set to zero above 5000 K.

Third, and recorded here because it will confuse anyone reading the paper against the code: **Frelikh's Equations (13) and (14) are printed without their logarithm**, as $E(800 < T < 1800\,\mathrm{K}) = -62.7016 + 0.0526104\,T - \ldots$ carrying units of W sr^{-1} molecule^{-1}. A negative polynomial cannot be a power, and the missing operator is a *natural* logarithm, not a common one. Two independent checks were run here. The two published segments disagree at their 1800 K junction by −2.35 % read as $\ln E$ and by −5.34 % read as $\log_{10} E$, and EXHALE's transcription of the same Table 5 records the mismatch as "2.4 % below": the natural-log value. And a two-level estimate for the ν_2 band ($A \approx 100$ s^{-1}, $E(\nu_2) = 3628$ K, $h\nu$ at 4 μm) gives 1.05×10^{-20} W sr^{-1} molecule^{-1} at 1000 K, against 3.34×10^{-20} for the natural-log reading and 1.4×10^{-45} for the common-log reading. Their own figures confirm that the code behind the paper is right and only the printed equations are wrong: their Figure 5 shows $\Lambda_{H_3^+} \approx 2 \times 10^{-7}$ erg cm^{-3} s^{-1} where Figure 4 shows $n_{H_3^+} \approx$ a few$\times 10^3$ cm^{-3}, which is the natural-log ratio.

6.6 Photoelectron secondary ionization

Section 5. records that EXHALE no longer treats the photoelectron heating fraction as a free parameter, which separates it from Taylor and Xing. Frelikh does not treat it as free either, and it uses a different and (for a molecular gas) a better prescription.

EXHALE uses **Shull & van Steenberg (1985)**, with the branching between the H I and He I secondary channels renormalized from the $n(\mathrm{He})/n(\mathrm{H}) = 0.1$ composition their Monte Carlo sampled onto the cell's own neutrals (`svs85_secondary_branching`, `radiation/util_ion_eq.f90`). Frelikh uses **Dalgarno, Yan & Liu (1999)**: $W = W_0(1 + Cx^\alpha)$ from their Tables 4 and 5, a secondary ionization added to the network as an extra reaction, and (this is the part EXHALE has no equivalent of) the H_2 channels put back explicitly. Their Equations (25)–(35) return the 5.4 eV dissociation heat following triplet excitation, the thermalized $\nu = 1$ (0.516 eV) and $\nu = 2$ (1.032 eV) vibrational levels with three separate ionization-fraction weightings, and the Lyman and Werner electronic states with their ∼90 % fluorescent / ∼10 % dissociative branching.

Frelikh also states the range of validity and checks it: "the approximations presented in A. Dalgarno et al. (1999) are only valid for a gas with an ionization fraction <0.1," which they justify by locating the relevant $\tau_\nu = 1$ surfaces in the neutral layer, with a stated error of at most 15 %.

The judgement is that for a molecular layer Dalgarno et al. (1999) is the correct prescription and Shull & van Steenberg (1985) is not. SvS85 is a Monte Carlo fit for a pure atomic H + He plasma; it carries no information about where photoelectron energy goes in a gas that is mostly H_2. Checked here, `svs85_secondary_branching` contains no H_2 term at all: the code carries an H_2 photoabsorption column separately, but the photoelectron energy partition has no H_2 vibrational thermalization, no dissociation heat return and no H_2 secondary ionization. The two prescriptions are complementary rather than competing: SvS85 remains valid to high ionization fraction, where Dalgarno's fits are not, and Dalgarno covers the molecular gas, where SvS85 has nothing to say. Neither covers the whole domain.

The sign of the effect is also worth recording, since it is not the intuitive one. Frelikh's Section 6 reports that including secondary ionization raised the ionization fraction in the molecular layer, increased electron-impact destruction of H_3^+,

reduced H_3^+ cooling and so *increased* the escape rate, "from 6.4×10^9 to 9×10^9 g s^{-1} sr^{-1} for our fiducial hot Jupiter run, which is counter to the intuition that secondary ionization results in a decreased efficiency of escape."

6.7 Metals and cooling

EXHALE has the richest *closed-form* metal treatment: ten elements (C, O, N, Mg, Si, Ca, Na, K, S, Fe) in 27 ion stages inside the coupled system, CHIANTI-fitted line cooling, ground-term fine-structure levels solved in statistical equilibrium at the local $(n_e, n_{\rm HI})$ rather than in the coronal limit, and line trapping in the eight fine-structure lines of C I, C II, N II and O I. Taylor can include solar-abundance metals but treats the structure as H/He-dominated. Xing has **no metals** and **only Lyα cooling**. Frelikh has no metals either, and only Lyα and H_3^+ cooling; it checked recombination cooling in postprocessing against the Osterbrock & Ferland rate and left it off as negligible. Its Section 5.4 names the omission as the main limit on extending the model to smaller planets, since H_3^+ is destroyed by H_2O and CO and metals bring line coolants of their own. SB23 itself carries only recombination and Lyα cooling; the AIOLOS code adds the Black (1981) rates with free-free and H_3^+ cooling, but neither is specialized to the metal-line escape problem.

6.8 Line diagnostics

Taylor computes 10830 and Hα in loop with the most complete non-LTE microphysics. Xing post-processes 10830 only. EXHALE builds the 2^3S and H(n=2) populations in the run (the metastable is on by default), and produces the transmission spectrum (10830, Lyα, Hα, Hβ, the Mg II, Ca II and Na I doublets) in `EXHALE_transit.py`, with impact-parameter Voigt integration, instrument and rotation convolution and an optional triaxial Roche geometry. AIOLOS does not target these lines.

6.9 Lower boundary

Taylor couples to a photochemical lower and middle atmosphere at $\sim$1 μbar, which is the most physically motivated boundary of the five. Xing uses a fixed-density, fixed-temperature base. AIOLOS leaves the boundary configurable. EXHALE now has the same range as Taylor: a fixed base, an analytic chemical-equilibrium column, a scalar handoff, or a full profile, item (D) below.

Frelikh imposes the same count as EXHALE, and one thing more. Its base sits at $P_0 = 0.96$ dyn cm^{-2}, i.e. at "the region of the outflow above the 1 μbar level, near which molecules such as H_2 are present," with $T_0 = 1000$ K and $R_{\rm p} = 10^{10}$ cm for the fiducial hot Jupiter (their Table 2). Their Section 2.8 states what is imposed and why:

> "At the inner boundary, the flow is subsonic, so we specify the boundary conditions for the number densities of each species and the pressure. We extrapolate the velocity from the grid: $u_0 = u_1 - (r_1 - r_0)(u_2 - u_1)/(r_2 - r_1)$, a wave can travel from the domain to the inner boundary, as the flow is subsonic, so the value of the velocity is part of the solution and cannot be imposed as a boundary condition."

EXHALE's `BC_component_constrho` pins ρ and p and copies the ghost velocity from the first interior cell. **The count of imposed conditions and the choice of which component floats are identical**, and Frelikh gives the characteristic

argument for that choice explicitly. **The difference is that Frelikh also imposes the composition.** That is the one item on EXHALE's own outstanding list for the base that Frelikh has already done; it appears as item (M) in Section 9..

Two further points. Frelikh nowhere reports a supersonic base, nor a guard against one. Searching the paper for that failure mode returns nothing: the subsonic condition is a *premise* of the Section 2.8 argument and the possibility of its being violated is not discussed. The structural weakness that follows (once the inflow is supersonic all three characteristics enter the domain and an extrapolated velocity stops being a boundary condition) applies to both codes; only EXHALE has found it.

Second, Frelikh's answer to the base thermal problem is not a physical model but a deliberate override, and the paper says so. Their Section 2.2 opens by naming the symptom: "We implement a bolometric heating and cooling term in this region, as the EUV heating is insufficient to prevent the gas from cooling to unphysically low temperatures", and then applies, below the $\tau = 3$ surface of the highest-energy bin, $\Gamma_{\rm bol} - \Lambda_{\rm IR} = \kappa\sigma_{\rm SB}(T_0^4 - T^4)$ with the opacity set by hand:

> "where we have set $\kappa_{\rm opt}$ and $\kappa_{\rm IR}$ to a single inflated value $\kappa = 1\,{\rm cm^2\,g^{-1}}$. In addition, in our implementation, we set $T_0^4 = 4T_{\rm eq}^4$, such that Equation (12) has the form of a Newtonian cooling term. … our choices effectively force the bolometrically heated layer to be isothermal at the temperature given by the input parameter T_0."

For the fiducial planet that region is $r < 1.005\,R_{\rm p}$. Heat conduction plays no part in it: Frelikh's Section 3.5 finds conduction "subdominant at all fluxes in our grid of models," worth $\sim$2 % in $\dot{M}$ at 0.3 au, and the only stability remark about it in the paper runs the other way, their Section 2.8 warns that "conduction makes the choice of outer boundary conditions more important, as the conduction term does not explicitly include a flux limiter."

The bearing on EXHALE is that EXHALE already owns a less blunt instrument for the same problem. `Base IR field` gives the H_3^+ band a radiative equilibrium fixed point at $T_{\rm rad} = 1140$ K, $W_{\rm dil} = 1/2$, i.e. 958.5 K on the MiZATeP cooling function (975.4 K with the Miller et al. 2013 fit used until 2026-10-10; header of `h3p_cooling.f90`), instead of letting it run the layer down; it is off by default (Section 6.5). The reservation is that the temperature trough measured in the molecular gate cases (before the cooling function changed; not re-measured since) reaches 865 K, *below* that fixed point, so H_3^+ alone does not account for it and the other channels in that region have not been separated.

6.10 The molecular layer, measured side by side

Frelikh's fiducial hot Jupiter and EXHALE's molecular regression cases are close enough in irradiation to compare directly. Frelikh quotes $F_{\rm EUV} = 2.6 \times 10^3$ erg cm^{-2} s^{-1} at 0.05 au; `EXHALE_setup.out` for the gate cases reports $\log_{10} F_{\rm XUV} = 3.193$, i.e. 1.56×10^3 erg cm^{-2} s^{-1}, a factor 1.7. The planets differ more: 0.7 $M_{\rm J}$ against a 14.52 $M_\oplus$ hot Uranus ($R_{\rm p} = 0.49\,R_{\rm J}$, $T_{\rm eq} = 1140$ K, $a = 0.048$ au, He/H = 0.0793).

All EXHALE numbers in Table 4 were measured here from `backup/regression/*/output/` by parsing the `# columns` headers; no case was re-run. Frelikh's numbers are from its Table 2 and its Figure 4, 5 and 9 captions, except the $\Lambda_{H_3^+}$ plateau, which was read off the Figure 5 axis and is therefore approximate.

Three readings of that table matter.

The three-layer structure reproduces, but on the wrong side of a qualitative divide. Frelikh's Section 5.4 sets its own super-Earth against Koskinen et al. (2022) and finds them alike, and both unlike a hot Jupiter:

Table 4: The molecular layer. EXHALE columns measured here; Frelikh columns as published.

	Frelikh HJ	**Frelikh s-Earth**	**EXHALE `mol_metals`**	**EXHALE `mol_base_handoff`**
Planet mass	$0.7\,M_{\rm J}$	$7.2\,M_\oplus$	$14.52\,M_\oplus$	$14.52\,M_\oplus$
Base pressure	$0.96\,\mu$bar	$0.47\,\mu$bar	$9.0\,\mu$bar	$9.0\,\mu$bar
Base T	1000 K (forced)	1000 K (forced)	1213 K	1213 K
Base H_2 fraction of H nuclei	≈ 1	≈ 1	0.99989	0.9996
$H_2 \rightarrow$H crossover	$1.01\,R_{\rm p}$	no sharp transition	$1.158\,R_{\rm p}$ (1166 K)	$1.158\,R_{\rm p}$ (1155 K)
Molecular layer cut-off	$1.04\,R_{\rm p}$, sharp	**none;** H_2 **survives throughout**	$1.19\,R_{\rm p}$, sharp	$1.19\,R_{\rm p}$, sharp
Molecular layer T	1000→4000 K	-	1213→1319 K	1213→1305 K
T minimum / maximum	- / ∼9500 K	- / 4200 K	**865 K** at 1.190 / 2719 K at 1.910	874 K / 2647 K
$n(H_3^+)$ maximum	$\sim 2\times10^5$ cm^{-3}	-	46.3 cm^{-3} at $1.124\,R_{\rm p}$	6.7×10^4 cm^{-3} at $1.031\,R_{\rm p}$
$N(H_3^+)$	1.0×10^{13} cm^{-2}	1.5×10^{13} cm^{-2}	1.73×10^{10} cm^{-2}	1.79×10^{13} cm^{-2}
$\Lambda_{H_3^+}$ maximum	$\sim 2\times10^{-7}$ (read off Fig. 5)	-	6.145×10^{-10} at $1.113\,R_{\rm p}$	8.898×10^{-7} at $1.031\,R_{\rm p}$
…as a fraction of local cooling	most of it	dominant	6.8 %	100 %
…as a multiple of local heating	$\lesssim 1$	-	0.003	**6.95**
Sonic point	$3.1\,R_{\rm p}$	$4.4\,R_{\rm p}$	$2.878\,R_{\rm p}$	$2.874\,R_{\rm p}$
Ionized fraction at sonic point	mostly ionized	mostly neutral	0.41	0.41
$\log_{10}\dot{M}$ [g s^{-1}]	11.05 if integrated over 4π	-	10.582	10.575
Peak Mach number below $1.35\,R_{\rm p}$	not reported	-	0.017	0.014

> "Our super-Earth runs are qualitatively similar to the results of their hot Uranus model, in which instead of the sharp transition between the molecular hydrogen layer and the atomic layer, as seen in our hot Jupiter simulations … molecules are present throughout the outflow. Particularly interesting is the presence of H_3^+ throughout the outflow, instead of being confined to a narrow region at the base."

EXHALE's hot Uranus gate does the opposite. Measured here, the H_2 fraction of H nuclei falls from 0.48 at $1.16\,R_p$ to 2.7×10^{-5} at $1.30\,R_p$, and $n(H_3^+)$ from 35 to 1.2×10^{-3} cm^{-3} over the same interval: four decades each. The temperature is also low: 865–2719 K, against the 4000–5000 K Frelikh quotes as the Koskinen hot Uranus prediction and the 1000–3800 K of its own super-Earth. **EXHALE's hot Uranus is behaving like Frelikh's hot Jupiter, not like the Koskinen hot Uranus the gate case is meant to reproduce.**

The two EXHALE gate cases disagree with each other by three orders of magnitude in $N(H_3^+)$, 1.73×10^{10} against 1.79×10^{13} cm^{-2}, and only the second is in Frelikh's range. The difference is the `base.inp` handoff, whose $q_{H_2,\mathrm{base}}$ sets the base H_3^+. In that case $\Lambda_{H_3^+}$ is 6.95 times the local heating at $1.031\,R_p$ and supplies 100 % of the cooling there, which is a net cooling excess visible in the numbers.

The base flow is not supersonic, and that is not the same as saying it is healthy. The peak Mach number below $1.35\,R_p$ is $1.4–1.7 \times 10^{-2}$ in every gate case, and the He/H of these runs (0.0793) is far outside the composition window where a supersonic base has been seen. What the same output does show is described in Section 8..

7. The H_2 thermal-dissociation pair: a defect diagnosed here, and since repaired on one side

Repaired on the EXHALE side, 2026-09-01; see Section 7.1. The diagnosis below is left as it was written, because it is what identified the defect and because it still holds for Frelikh. The EXHALE column of Table 5 describes the code as it was before that date.

Frelikh's k12 and k13 are the forward and reverse of the same process, H_2+M $\rightleftharpoons$ H+H+M, taken from two different measurements: $1.5 \times 10^{-9} e^{-4.8\times10^4/T}$ from Baulch et al. (1992) for the dissociation and $8.0 \times 10^{-33}(300/T)^{0.6}$ from Ham et al. (1970) for the three-body recombination. EXHALE's R12 and R15 were the same pair from the same two sources, differing only in that Koskinen et al. (2022) print the barrier as 48 350 K where Frelikh rounds it to 4.8×10^4 K.

Nothing in Frelikh discusses whether the pair is thermodynamically consistent. The paper was searched here for *detailed balance*, *thermodynamic*, *equilibrium constant*, *reverse reaction*, *reversible*, *extrapolat* and *range of validity*, across the body, the appendices, the footnotes and the table notes. The only hits are the linear velocity extrapolation of Section 2.8 and the Bulirsch–Stoer extrapolation of Appendix B, both unrelated. **There is no statement about the consistency of the pair, and none about the temperature range over which either fit is valid.** No such record was found on the EXHALE side either; the caveats in `mol_rates.f90` concern third-body efficiency and the helium-rich limit, not forward–reverse consistency.

The ratio $k_{\mathrm{diss}}/k_{\mathrm{rec}}$ is the equilibrium constant n_H^2/n_{H_2}, with the third body cancelling, so it can be checked against thermodynamics directly. Table 5 does that.[1]

[1] $K = (2\pi\mu k_B T/h^2)^{3/2} g_H^2/z_{\mathrm{int}}(H_2)\; e^{-D_0/k_B T}$ with $\mu = m_H/2$, $g_H = 2$ and $D_0 = 36\,118.11$ cm^{-1}. The internal partition function was summed explicitly over v = 0–15 and J = 0–79 with the Huber & Herzberg constants (ω_e = 4401.21, $\omega_e x_e$ = 121.33, $\omega_e y_e$ = 0.812, B_e = 60.853, α_e =

Table 5: The thermal-dissociation pair against thermodynamic equilibrium. Ratios above unity mean the pair drives the gas more atomic than equilibrium allows. Computed here. † EXHALE as it was before 2026-09-01; Section 7.1.

T [K]	K_{thermo} [cm^{-3}]	Frelikh k_{12}/k_{13}	ratio	EXHALE R12/R15†	ratio
500	7.07×10^{-22}	5.17×10^{-19}	**732**	2.57×10^{-19}	**363**
800	7.71×10^{-5}	2.96×10^{-3}	**38.4**	1.91×10^{-3}	**24.8**
1000	3.80×10^{1}	5.50×10^{2}	**14.5**	3.88×10^{2}	**10.2**
1500	1.54×10^{9}	6.24×10^{9}	4.05	4.94×10^{9}	3.21
2000	9.96×10^{12}	2.21×10^{13}	2.22	1.86×10^{13}	1.86
3000	6.38×10^{16}	8.40×10^{16}	1.32	7.48×10^{16}	1.17
4000	4.97×10^{18}	5.45×10^{18}	1.10	4.97×10^{18}	1.00
5000	6.62×10^{19}	6.87×10^{19}	1.04	6.40×10^{19}	0.97

The judgement is that the pair is thermodynamically consistent only above about 3000–4000 K, and that this is a defect of both codes equally. Below that it drives the composition toward atomic hydrogen faster than equilibrium permits, by a factor 14.5 in Frelikh and 10.2 in EXHALE at 1000 K, and by 38 and 25 at 800 K. EXHALE is the less wrong of the two only because the higher barrier it transcribes partly compensates; the error is the same error. Both codes place their molecular layer in exactly the 1000–1500 K band where the inconsistency is largest: Frelikh forces its base to $T_0 = 1000$ K, and the EXHALE gate cases measure 1213–1319 K through the layer.

Two qualifications. Neither code sets the H_2/H balance from this pair alone: photodissociation, the Lyman–Werner channel and the ion chemistry all act, so the composition error is smaller than the tabulated factors and is confined to wherever thermal dissociation and three-body recombination dominate. And this is a statement about the pair, not about either published measurement in its own range of validity; the Baulch fit belongs to shock-tube temperatures far above the molecular base.

7.1 The repair, 2026-09-01

EXHALE no longer transcribes the dissociation fit. `rk_R12_H2_thdis` in `src/modules/lower_atmosphere/mol_rates.f90` now returns `k3b_H_H_to_H2(T) / keq_H_H_to_H2(T)`: the three-body recombination coefficient divided by the equilibrium constant the module builds from the H_2 spectroscopic constants. R15 moved at the same time from the Ham et al. (1970) room-temperature measurement to the Cohen & Westberg (1983) recommended $k_1(H_2) = 2.8 \times 10^{-31} T^{-0.6}$, whose stated 50–5000 K range covers the molecular layer where Ham's 77–300 K does not. The consequence for Table 5 is that the EXHALE ratio column becomes **1 by construction at every temperature**: R12/R15 is now $1/K_{\mathrm{eq}}$ exactly, and against the independently computed K_{thermo} of the footnote it is 0.998–1.000 over 500–5000 K, the residual being the two evaluations' different treatment of the H_2 internal partition function. The 363, 24.8, 10.2 and 3.21 of the 500–1500 K rows are gone.

Two checks that the construction is right rather than merely self-consistent. Against the Baulch fit it replaces, the new k_{diss} is 1.05 at 8000 K, 1.18 at 5000 K, 0.97 at 3000 K and 0.82 at 2500 K (agreement inside Baulch's own stated 2500–8000 K range), and departs only below it (0.61 at 2000 K, 0.11 at 1000 K), which is where the fit is an extrapolation.

3.062 cm^{-1}), discarding levels above D_0, with ortho/para nuclear-spin weights 3 and 1 divided out by $(2I+1)^2 = 4$ so that the nuclear spin cancels consistently against the H atoms. A rigid-rotor harmonic-oscillator evaluation of the same expression agrees to 2.8 % at 1000 K and 1.7 % at 2000 K.

Against the primary shock-tube measurement rather than the evaluation, Breshears & Bird's own M = H_2 coefficient $5.48 \times 10^{-9} e^{-52989/T}$, the ratio is 1.11 at 3500 K (the bottom of their measured range), 0.99 at 4000 K, 0.82 at 5000 K and 0.51 at 8000 K.

The judgement of this section therefore stands for Frelikh and no longer for EXHALE. The pair is now thermodynamically exact in EXHALE at every temperature, and its molecular layer is no longer driven toward atomic hydrogen faster than equilibrium permits. What the detailed-balance route assumes instead is the vibrational state the coefficients refer to, which is recorded at the code site and in `docs/molecular_hydrogen_treatment.tex`.

8. What the molecular gate cases show

The comparison in Section 6.10 required reading EXHALE's molecular regression output closely, and five things turned up in it that are not about Frelikh at all. They are recorded here with the method, so that they can be re-measured. Every one comes from `backup/regression/<case>/` with no case re-run; the profile quantities are from `output/Hydro_ioniz.txt` columns r/R_p, ρ [m_H cm^{-3}] and v, and the warnings from `run.log`. All five gate cases are relaxation snapshots pinned at `maxsteps` = 12000.

1. **The mass flux is not flat at the base.** $\rho v r^2$ in `mol_metals` runs 1.6563×10^{16} at $r = 1.00019$ and 3.1420×10^{16} at $r = 1.00039$: a factor **1.90 across one cell**, at the second interior cell. Over the molecular layer ($1.0 < r/R_p < 1.19$) the spread (max − min)/mean is **1.865**. In `mol_lyman_werner` the first two rows are exactly zero, the third is **negative**, -1.6507×10^{16}, and the fourth returns to $+1.3357 \times 10^{15}$: the flux changes sign inside the layer. A converged steady state requires $\rho v r^2$ to be constant in radius, and these are not.
2. **Inflowing cells sit inside the molecular layer in one case.** Counting rows with $v < 0$: `mol_metals`, `mol_base_handoff` and `mol_diffusion` each have 28, all at r = 1.305–1.426 R_p, i.e. *outside* the molecular front at 1.16–1.19. `mol_lyman_werner` has **144, at** r = 1.00019**–**1.12394 R_p**, inside the layer**, with the first cell's velocity exactly zero: the one-way valve latched shut.
3. **The accepted chemical states that are not roots sit at the temperature minimum.** `mol_metals/run.log` carries 24 lines of `(ioniz_eq) NON-ROOT accepted (relaxation amnesty)` and `mol_ir_bands` 11; an example reads `cell 260 step 1 r 1.197620 T[K] 1.197E+03 info 4 viol 1.210E-08 res 1.065E-03`. They cluster at $r \approx 1.19$–$1.20\,R_p$, which is where the temperature trough bottoms out at 865 K (Table 4). **The direction of causation was not determined.** Whether a runaway in the cooling breaks the chemical solve, or a state that is not a root drives the cooling, or a third cause produces both, is not decided by this measurement, and no inference is drawn here. What the coincidence does put in question is whether the thermal collapse and the non-uniqueness of the chemical solve are two problems or one.
4. **Lyman–Werner self-shielding is being evaluated outside its fit.** `mol_lyman_werner/run.log`: "`the H2 Lyman-Werner lines remove 1.01E+00 of the 912-1110 A band, i.e. all of it. The Draine & Bertoldi (1996) self-shielding fit is a fit to a RATE and does not bound the photon budget; the band transmission is floored at zero and this column is outside the fit.`" The band is 101 % removed and the transmission floors at zero. This is the same case as item 2; no connection between the two is asserted.

5. **`mol_ir_bands` exercises one of the three channels its name implies.** With `Base IR field: True` and `Molecular IR bands: True`, the H_2O and CO infrared columns of `output/Cooling_breakdown.txt` (col. 13 and 14) are **exactly zero at every radius**, because `Oxygen chemistry` is off in that configuration and the carriers do not exist. The only molecular infrared channel actually switched on is the H_2 band, at 8.98×10^{-9} at the H_3^+ peak cell and 2.93×10^{-7} erg cm^{-3} s^{-1} at its own maximum. As a regression gate the case does not protect the H_2O and CO paths.

A note on quoting $\dot{M}$ from these cases, since it is easy to get wrong. `src/EXHALE_main.f90` evaluates it at $j = N - 20$, i.e. at $r = 3.9231\,R_p$, and halves it under `2D approximate method: Rate/2 + Mdot/2`. Recomputing $4\pi\rho m_H v r^2$ at that index and halving reproduces the $\log_{10}\dot{M} = 10.58$ that `EXHALE_setup.out` prints. Evaluating at the outer edge instead, or omitting the factor of two, gives 10.94 and is not the quantity the code reports. In any case the mass-flux spread above means none of these numbers should be quoted as a converged $\dot{M}$.

For completeness: the temperature floor these cases could have hit is $0.01\,T_0 = 11.4$ K, because `T_trial` in `energy_semi_implicit.f90` is dimensionless. No case activated it (`grep -c floor` over `run.log` returns zero in all five), and the measured minima (865, 874, 946 and 970 K) differ case by case rather than resting on a common constant.

9. Adoption assessment

Each candidate is given one of three verdicts. **Adopted** names the module that implements it. **Adopted then replaced** records that the element entered the code and was later superseded, and by what. **Not adopted** states whether the original reason still holds.

9.1 Adopted

(A) He/H diffusive separation: adopted, in a stronger form than recommended. The recommendation was Taylor's route: a molecular-plus-eddy diffusion term on top of a single bulk momentum solve. That is what `functions/binary_element_diffusion.f90` does, with the two departures described in Section 6.1, the transported variable is the helium mass fraction of a binary mixture rather than a trace ratio, and the binary coefficient is resolved by ionization stage rather than taken at a neutral rate. The eddy term comes from the scalar key `He_Kzz` or, when a lower-atmosphere profile is in use, from that profile's $K_{zz}(p)$ interpolated onto the grid. One implicit backward-Euler step is taken per hydro step and converged by Newton on a tridiagonal Jacobian, with a Péclet-switched donor-cell treatment of the $X(1-X)$ drift product so the flux vanishes where either component runs out. Metals ride with hydrogen at fixed metal/H unless `He_metal_diffusion` is set, in which case each metal additionally diffuses as a trace species through the solved hydrogen background. The operator is available both inside the marching loop and, through an outer composition relaxation, on the direct steady route. Default off (`He_diffusion`).

(D) Whole-atmosphere lower boundary: adopted, and carried past Taylor. EXHALE now has three routes below the base, in increasing order of what they carry. An analytic isothermal column from the 1-bar level to the μbar base using the Koskinen et al. (2022, Eqs. 11–13) chemical-equilibrium H_2/H/He partition

(`lower_atmosphere/lower_column.f90`). A scalar handoff file `base.inp`, generated analytically or from a VULCAN run, which carries the EOS boundary (T, r, p, q_{H_2}), the elemental reservoirs (He/H and a `<El>_H_base` key for each of the ten elements, which is what `melem_ab` is then built from) and the eddy coefficient. And a full *profile* named by `Lower atmosphere profile:` (`files_IO/lower_atmosphere_profile.f90`), which carries p, r, T, n_{tot}, ρ, K_{zz}, q_{H_2}, q_H, X_{He} and a mixing-ratio column per element, interpolated in $\log p$. A scalar physics key sitting beside a profile is refused rather than merged, and the two files must agree on a `solution_id`.

The part that goes beyond Taylor is the closure. Taylor's coupling is one-way: the photochemistry sets the base and the wind reads it. EXHALE closes the loop on the elemental fluxes. `binary_element_diffusion.f90` measures F_H and F_{He} face by face and reduces them to a median and a spread over a radial window, and `src/utils/element_flux_closure.py` iterates $\Phi_{El} = F_{El}(\Phi_{El})$ to a fixed point with under-relaxation, the chemistry taking Φ as its upper boundary condition and the wind returning F. The elemental composition of the wind is then an output of the coupled pair rather than an input to either. `photochem_to_lower_profile.py` and `vulcan_to_lower_profile.py` produce the profile file; a residual smaller than the spread of the window it was measured on is reported as unresolved rather than as converged.

(E) Molecular chemistry and H_3^+ cooling: adopted. H_2, H_2^+, H_3^+ and HeH^+ are unknowns of the same coupled system as H, He and the metals (`System_HeH_mol.f90`, `System_HeH_mol_metals.f90`), so the molecular and metal blocks share one electron density. H_3^+ infrared cooling is the MiZATeP LTE cooling function (Mizus et al. 2017; the Miller et al. 2013 fit until 2026-10-10) with the Miller et al. (2013) non-LTE departure factor and with absorption of the ambient infrared field so the rate returned is the net one (`lower_atmosphere/h3p_cooling.f90`). H_2 photodissociation in the Lyman–Werner bands uses Draine & Bertoldi (1996) self-shielding (`lower_atmosphere/lyman_werner.f90`, key `Stellar LW flux`, zero by default). The three-body density M of the association reactions is the total gas-particle density from `calc_ntot`, not a mass density. Default off (`Molecular chemistry`); turning it on requires helium and forces the molecular particle count at the base.

(F) Lyα Monte Carlo: adopted as an option, not as the default. The recommendation was to keep the escape-probability closure and use Taylor's Monte Carlo only as a validation benchmark, because a WASP-121 b column with $\tau_0 = 1.2 \times 10^8$ makes an acceleration-free Monte Carlo intractable and forbids core skipping. That still stands for the in-loop field: the default is the in-line static-slab closure. But EXHALE also accepts a tabulated $J_{Ly\alpha}(r)$ from any external solver (`Jlya RT file`), and that path is used with the LaRT Monte Carlo code, including the Lyα emitted in situ within the wind. The Monte Carlo is therefore available where it is affordable, without being in the iteration.

(I) Navier–Stokes viscosity and heat conduction: adopted (from Koskinen, i.e. Taylor's own hydro heritage). This was not on the original list. `time_step/viscous_conduction.f90` adds the viscous momentum force, its dissipation and thermal conduction (the Navier–Stokes level that CETIMB solves and that an inviscid HLLC scheme lacks) integrated Crank–Nicolson and entering the steady residual through the same operator. Both are opt-in (`Viscosity`, `Conduction`). Note that the module does *not* implement Koskinen et al. (2022) Eqs. (B5) and (B6) as printed: for a

radial flow with the Newtonian stress $\tau_{ij} = \mu(\partial_i w_j + \partial_j w_i) - \frac{2}{3}\mu\delta_{ij}\nabla\cdot\mathbf{w}$, the viscous force and the dissipation are

$$F_\mu = \frac{4}{3}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\mu\frac{\partial w}{\partial r}\right) - \frac{4}{3}\frac{\partial\mu}{\partial r}\frac{w}{r} - \frac{8}{3}\frac{\mu w}{r^2}, \qquad q_\mu = \frac{4}{3}\mu\left(\frac{\partial w}{\partial r} - \frac{w}{r}\right)^2,$$

and the code carries these rather than the printed forms; the derivation is in the module header.

(J) AIOLOS's opacity-model dispatcher: adopted. Also not on the original list, which treated AIOLOS as a cross-check only. `radiation/opacity_models.f90` is an AIOLOS-style dispatcher over photoionization opacity models (analytic, constant, constant with Robinson & Catling pressure broadening, and tabulated from a two-column `.opa` file) selected at runtime by the presence of `opacity.inp`. The pressure factor that was a deferred hook in the version it came through is applied cell by cell here.

9.2 Adopted, then replaced

(B) Temperature-dependent Penning ionization: adopted from Taylor (2025), then retired. The original recommendation was the cheapest item on the list, and it was taken: the constant 5×10^{-10} cm^3 s^{-1} of Roberge & Dalgarno (1982) was replaced by the two-branch power law of Taylor et al. (2025) Table 2, unconditionally. It has since been withdrawn. The fit was transcribed correctly but is not usable: it **jumps by a factor 1.5724 at 4000 K** ($1.5849 \times 10^{-9} \rightarrow 2.4921 \times 10^{-9}$), and a rate coefficient is a continuous function of temperature. It also disagrees with its own paper's Figure 19, where the Maxwell–Boltzmann average of the Cohen & Lane (1971) and Morgner & Niehaus (1979) cross sections rises to 1.50×10^{-9} near 1400 K and falls to 1.29×10^{-9} at 4000 K while the tabulated branch decreases monotonically, and above 4000 K it exceeds every cross-section determination collected by García Muñoz (2025).

What is in the code now is the continuous closed form of García Muñoz (2025), *A&A* 698, A199: the Maxwell–Boltzmann average of the Movre & Meyer (1997) total ionization cross sections for the atomic partner (`ioniz_HeI23S_H`) and of Cohen & Lane (1977) for the molecular one (`ioniz_HeI23S_H2`), both in `radiation/Cool_coeff.f90`. Both return the *total* rate that removes the metastable; the 0.9/0.1 split between the Penning channel, which leaves a lasting ion and a free electron, and the associative channel, which makes HeH^+, is carried by `f_penning_HeI23S` at the points where the products matter. Taylor's own Figure 19 agrees with the adopted expression to 10–20 %, so the two independent calculations are consistent and it was the published fit that was the outlier.

(K) AIOLOS's C/N/O line-cooling fits: adopted, then replaced by CHIANTI. The analytic C/N/O metal-cooling fits of AIOLOS's `photochem.cpp` (code, not paper) were ported into `Cool_coeff.f90` and were the default for a time. They are now the legacy branch (`cno_cool 0`) and the default is a set of closed-form fits to CHIANTI, accurate to 0.2–2.1 % over 10^3–10^5 K. The reason is physical, not stylistic: the AIOLOS fits contain no nitrogen cooling at all, and their O I/O II rates deviate from CHIANTI by 40–70 %. The AIOLOS branch is kept so old results remain reproducible, and its remaining tuning constants are flagged in place.

9.3 Not adopted

(C) The B-spline K-matrix He(2^3S) cross section, not adopted; the reason still holds, but the earlier statement of it was too generous. It was claimed that EXHALE was "already most of the way there." What `functions/cross_sec.f90` actually has is a smooth background: two Verner et al. (1996) single-shell forms joined by a log-linear bridge, fitted at the few-percent level to the Norcross (1971) / TOPbase background, with the autoionization region represented by a *resonance-averaged* bump. Taylor's cross section resolves the individual EUV resonances, which EXHALE's does not and cannot. The case for adopting it is unchanged: low cost, a data swap; moderate value, since whether the resonances matter depends on where the star's coronal lines fall, but it should be stated as a missing capability rather than as a finished job.

(G) AIOLOS's FLD radiation and general reaction network, not adopted; the reason still holds, with one clause of it withdrawn. FLD matters for optically-thick interiors and dust, not for an optically-thin thermosphere, and a general implicit reaction matrix is the wrong shape for a steady-state code whose ionization system is one algebraic solve. Both still stand. The third clause of the old argument (that HLLC would be no improvement on EXHALE's "approximate Riemann solver") was simply wrong about the code: EXHALE has had HLLC since ATES, alongside LLF and Roe, and HLLC is what every example configuration uses. There was nothing to adopt there.

(H) Xing's electron-pressure Riemann split and full multi-fluid, not adopted; the reason is now stronger. Converting EXHALE to genuine multi-fluid means velocities per species, collisional drag and the electron-pressure split, which is a rewrite of the hydro core and hard to reconcile with a single coupled algebraic ionization solve. The original argument was that Taylor's diffusion approximation captures most of the same effect far more cheaply. That argument is now backed by what was built: the stage-resolved binary operator reproduces the friction physics (an ion held to the protons by Coulomb collisions, a neutral settling at a hard-sphere rate) that motivated the multi-fluid treatment in the first place. Xing remains a validation reference for the fractionation itself.

(L) The Dalgarno et al. (1999) photoelectron partition in a molecular gas, not adopted; this one is a real gap. Unlike (C), (G) and (H), the reason here does *not* still hold. EXHALE uses Shull & van Steenberg (1985) throughout, and its branching routine contains no H_2 term of any kind (Section 6.6): no vibrational thermalization, no dissociation heat returned, no H_2 secondary ionization. That was harmless while the code had no molecular layer. It is not harmless now that `Molecular chemistry` exists, because the layer it omits the physics from is exactly the layer where the highest-energy photons deposit. The prescriptions are complementary (SvS85 stays valid where Dalgarno's fits do not, above an ionization fraction of 0.1), so the shape of the fix is a composition switch between the two rather than a replacement. Cost: moderate, a set of closed-form fits from Dalgarno et al. (1999) Tables 4 and 5 plus their Equations (25)–(35). Value: it changes the energy budget of the molecular layer, and Frelikh reports the effect running counter to intuition, raising $\dot{M}$ rather than lowering it.

(M) A composition boundary condition at the base, not adopted; recommended. Frelikh imposes the number density of every species at the inner boundary along with the pressure (Section 6.9); EXHALE pins ρ and p and lets the

interior chemistry decide what the gas at the base is made of. The imposed-condition count is the same either way, so this is not a well-posedness question: it is a question of whether an inflowing molecular reservoir is allowed to arrive with a composition of its own. EXHALE's own outstanding list already carries this item; the point worth recording is that a peer code does it and reports no difficulty from it.

(N) Time integration of the network in molecular cells, not adopted; recorded as an option rather than a recommendation. The bistability that forces the radiation-field continuation in `constrained_chemical_equilibrium.f90` is a cost of solving for equilibrium, and a short time integration of the same network does not have it (Section 6.3). Frelikh's stepper is an off-the-shelf semi-implicit Euler with an analytic Jacobian, which is not expensive. Against that: EXHALE is a steady-state code, its molecular rows share one electron density with the metal and He(2^3S) rows, and splitting the molecular block out into a time-integrated substep would break the single-system architecture that the rest of this memo records as the right one. The honest statement is that the continuation and the substep solve the same difficulty and neither has been measured against the other here.

(O) Post-hoc verification of the radiative-transfer approximations, not adopted; cheap, and worth doing. Frelikh does two checks that EXHALE asserts rather than measures. Their Appendix D runs a Monte Carlo of 1000 Lyα photons through the converged profile and reports 78 % escaping to space, 11 % lost downward into the bolometric region and 11 % thermalized, which is what licenses the optically thin Lyα cooling rate; repeating it from deeper in the molecular layer gives 36/27/37 and they say so. Their Section 3.4 checks H_3^+ optical thinness against ExoCross cross sections at the computed column. EXHALE has the ingredients for both (the LaRT coupling for the first, the H_3^+ cooling data for the second), and uses neither as a check on its own closures.

10. What was not verified

The Frelikh comparison rests in places on sources that were not available, and on EXHALE paths that were not followed to the end. Six items are open, and are listed so that nothing above is read as more settled than it is.

1. **Yelle (2004) is not in `references/`**, so whether Frelikh's barrier-free H_3^++H constant faithfully transcribes its stated source could not be checked. The thermochemical argument in Section 6.4 stands on its own, but it convicts the printed rate, not necessarily the transcription.
2. **Larsson et al. (2008) and Baulch et al. (1992) are likewise not held**, so the H_3^+ recombination factor of 1.60 and the 48 350 versus 4.8×10^4 K barrier were compared between the two secondary sources, not against the primary measurements.
3. **Whether EXHALE implements the dissociative photoionization channel** $H_2+h\nu \rightarrow H+H^+ + e^-$ was not established. `PIR_H2` exists in `util_ion_eq.f90`; the branching among the H_2 photo-channels was not traced.
4. **The temperature clamp reported elsewhere was not reproduced.** The five molecular gate cases do not activate the 11.4 K floor, and their minima differ case by case (Section 8.). Whatever run exhibited a clamped cell, it is not one of these.
5. **The direction of causation between the temperature trough and the non-root chemical states** at $r \approx 1.19\,R_p$ is

undetermined (Section 8., item 3).

6. **Frelikh contradicts itself on the number of evolved species**, and the memo does not adjudicate. Its Section 2.9 says "the seven atomic/molecular species (+electrons)"; its Appendix A says "the eight component species"; its Table 1 contains eight neutral and ionic species (H, H^+, H_2, H_2^+, H_3^+, He, He^+, HeH^+).

One further observation is recorded as an observation only. In Frelikh's own molecular layer, reading $n(\mathrm{H}) \approx 10^{11}$ and $n_e \approx 3 \times 10^8$ cm^{-3} off its Figure 4, the barrier-free H_3^++H channel gives ≈ 200 s^{-1} against ≈ 12 s^{-1} for dissociative recombination, which would make it the dominant H_3^+ sink. That sits awkwardly against their Section 2.3, which says of the recombination channels that "we verify post facto that they do indeed dominate in our results." The numbers behind this were read off a published figure, so it is not asserted as a finding.

11. Summary

- **AIOLOS** is a general-purpose, multi-fluid RHD code with the most sophisticated *numerics* and the least specialization to the He/Hα escape-diagnostic problem. It has contributed less than Taylor to EXHALE, but not nothing: the opacity-model dispatcher came from it, and its C/N/O cooling fits were EXHALE's default until CHIANTI replaced them.
- **Taylor (2025/2026)** was and remains the closest match to EXHALE's goals and its richest source of upgrades. Four of the six adopted items come from Taylor or from the Koskinen code beneath it: diffusive He/H separation, molecular chemistry with H_3^+ cooling, the whole-atmosphere lower boundary, and Navier–Stokes viscosity and conduction. The one Taylor item that did not survive is the Penning rate, and it failed on its own internal inconsistency rather than on anything about EXHALE.
- **Xing (2023)** still provides the physically purest treatment of He/H mass fractionation, and is still not something to import. It is a validation reference.
- **Frelikh (2026)** is the closest peer to EXHALE's molecular layer and, unlike the other three, it is a code EXHALE can be checked *against* rather than borrowed from: twelve of its 22 reactions are identical to EXHALE's in value and source, and where the two differ EXHALE is ahead on the network (it has the He↔H charge exchange Frelikh lacks, and the endothermic barrier on H_3^++H that Frelikh's constant rate is missing), ahead on H_3^+ cooling (non-LTE departure factor, four temperature segments, an optional exchange with the base field), and behind on exactly two things: the photoelectron partition in a molecular gas, and a composition boundary condition at the base.
- **One finding belonged to no single code, and has since been repaired on one side.** The thermal-dissociation pair both codes took from Baulch (1992) and Ham (1970) is off thermodynamic equilibrium by a factor 10–15 at 1000 K in the sense of over-dissociating (Section 7.); EXHALE now builds the dissociation from the recombination through the equilibrium constant and is exact (Section 7.1), and the finding stands for Frelikh only. The extrapolated-velocity inner boundary both codes use stops being a boundary condition if the inflow ever goes supersonic, a weakness only EXHALE has so far found in itself, belongs to both.
- **The two gaps this memo was written to name are closed.** EXHALE has radius-dependent He/H separation, and it has a lower boundary that a photochemical model sets rather than the user. What is left of the original list

is one data swap (the resolved He(2^3S) cross section) and two architectural items that were correctly declined. The Frelikh comparison has since opened four more, of which one (the missing H_2 channels in the photoelectron partition) is a genuine gap rather than a declined option.